Paper 14 — Measuring the Semantic Residual: The Text Limit and Grounding Fill-In

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Abstract

This paper turns "the semantic residual" from a vague phrase into a measurable quantity.

First, let the definition be written clearly: language is humanity's compression of the world, and language can be written as p-language + R-language; p-language is the part that can be approximated by functions, while R-language is only declared, never unfolded. Semantics is the alignment relation between language and phenomena; the semantic residual = the error floor of this alignment mapping under a given modality combination.

The experiment uses controlled ambiguity: the text describes the same synthetic object in three ways — precise words, vague grouped words, and omission — and the ambiguity is generated by the program, so "the limit of what the text can give" can be computed analytically (the Bayes floor). Then three quantities are measured: the Bayes floor, the error of a text-only model, and the error of the model after text plus image (grounding).

Results: the text model's error differs from the Bayes floor by only 1.5 to 2.7 percentage points (0.725 vs 0.750; 0.835 vs 0.850; 0.688 vs 0.714), showing that the fit is already in place and that the remaining residual comes from the text itself; after grounding, the accuracy on all three attributes reaches 1.000, showing that the dimension the text lacks is supplied by the image.

Conclusion: the semantic residual can be defined and can be measured; it divides into "the irreducible floor under that modality" (endogenous to ambiguity) and "the reducible part" (what grounding/added data can eliminate). This distinction gives "how much semantics has AI understood" a denominator.

I. The Problem

We have said two things before: language is humanity's compression of the world; what large models compress is the p-language of language.

But between "p-language" and "semantics" there is still a missing measurable link. The question this paper wants to answer is: can "the semantic residual" be defined as a number, and measured with a toy experiment?

To do this, three things are needed: a definition (what the semantic residual is), a reference line (how much limit the text itself can give), and a control (whether the residual drops once another modality is added).

II. Definitions

2.1 Definitions carried over from the series

Universe = P + R, where P is the functionizable part and R is the non-functionizable part. Language is likewise written as p-language + R-language: p-language can be approximated by functions, while R-language is only declared, never unfolded. Life = p-life + R-life, where p-life is the dynamic function class that lies in P and is preserved by the universe.

This paper deals only with the p-language side.

2.2 Semantics is a mapping

Define semantics as a mapping: given a language expression and the phenomenon it corresponds to (an image, an event, a state, an action, a quantity), semantics is the correspondence "expression → phenomenon".

This mapping is not single-layer: it has several dimensions by modality — the visual dimension (words to images), the event dimension (sentences to timelines), the state dimension (descriptions to system states), the action dimension (instructions to consequences), the quantitative dimension (degree words to measurable quantities). The semantic residual appears by dimension.

2.3 The semantic residual

Definition (semantic residual): Fix a modality combination M (for example: text only; or text plus image), and let the error metric be the error rate in predicting the attributes of the phenomenon. The semantic residual = the minimum error that any function can reach under the modality combination M.

This definition splits the residual into two parts:

- **The irreducible floor**: the error floor determined by the modality combination itself. When the text is ambiguous, some information is simply not in the text, and no function can fill it in.

- **The reducible part**: the gap caused by the model not having fitted well enough; it can be lowered by adding data, capacity, or modalities.

The criterion is direct: after adding a means, does the error floor drop or not. Drop ⇒ reducible; no drop ⇒ the irreducible floor under that modality.

2.4 The reference line: the Bayes floor

In the toy world the text generation process is known, so "the limit the text can give" can be computed analytically: given a piece of text, the posterior distribution over attributes is determined by the generation process; the accuracy of the Bayes-optimal predictor is the upper bound.

This upper bound is the "text limit". No model can be higher than it; if a model can approach it, that shows the fit is in place.

III. Experiment Design

World: synthetic objects with three attributes — color (4 kinds), shape (2 kinds), count (1 to 3). Objects are rendered into 8×8 images (color, shape, and count are all visible).

Text: a description assembled from three slots, each slot having controlled ambiguity.

- Color slot: with probability 0.5 a precise word (red/blue/green/yellow), with probability 0.5 a vague grouped word (warm colors = red or yellow; cool colors = blue or green).

- Shape slot: with probability 0.7 a precise word (circle/square), with probability 0.3 omitted.

- Count slot: with probability 0.5 a precise word (one/two/three), with probability 0.5 a vague word (a few = one or two).

Because the generation process is known, the posterior over attributes corresponding to each piece of text can be computed, and the Bayes floor can therefore be computed analytically.

Model: a small two-layer network (hidden layer 32, about 2000 parameters). The text model sees only the text plus a task identifier; the grounded model also sees the image (8×8×3, flattened).

Three quantities: the Bayes floor, the text model's accuracy, the grounded model's accuracy. The difference is the "reducible residual".

IV. Results

The Bayes floor for color is 0.750, the text model 0.725, the grounded model 1.000. The Bayes floor for shape is 0.850, the text model 0.835, the grounded model 1.000. The Bayes floor for count is 0.714, the text model 0.688, the grounded model 1.000.

Three points of interpretation.

First, the text model differs from the Bayes floor by only 1.5 to 2.7 percentage points. This shows that the model's fit to the text is already in place; the remaining gap is not that the model is inadequate, but that the text simply does not contain that information.

Second, after grounding all three attributes reach 1.000. The image supplies the dimension the text lacks, and the residual is filled in. So the part of the residual under the "text limit" belongs to the modality combination, not to the world: change the modality combination and it can be eliminated.

Third, the semantic residual obtained this way is measurable layer by layer: the text-only residual (around 25% for color, 15% for shape, 29% for count) is "the limit the text can give"; the residual after grounding is 0, showing that this part of the residual is reducible, only it needs another modality.

V. What It Shows and What It Does Not Show

Shows: the semantic residual can be defined as a number; that number can be split into "the irreducible floor" and "the reducible part"; in the toy world the boundary between the two can be computed analytically and matches the model's behaviour.

Does not show: it does not prove how large the semantic residual of real language is; it does not prove that R-language exists or does not exist (this paper deals only with the p-language side); it does not prove that "grounding can always fill things in" holds on all tasks — this paper measures only the visual dimension, and other dimensions (event, state, action) are not covered.

VI. Limitations

First, the world is synthetic and the ambiguity is constructed by a program; the ambiguity of real language comes from history and usage, and its form is different.

Second, the model is small (about 2000 parameters), not a large model; but what this paper measures is the mechanism of "text limit vs grounding fill-in", which is independent of scale.

Third, only one modality of grounding (images) is done; the residual under multiple modalities (events, actions) is not measured.

Fourth, the Bayes floor is computable only when the generation process is known; in real language this floor has to be approximated with the human disagreement rate (the next piece of work).

VII. Relation to Existing Work

Measuring the semantic residual is not a new topic; there are at least three existing threads.

First, human label variation: for the same piece of text, different annotators will naturally give different interpretations, and "the human ceiling" and "annotation disagreement" have been reported all along in evaluation.

Second, ambiguity benchmarks: tasks such as AmbigQA and word-sense disambiguation specifically test the case where "the same sentence has several reasonable readings".

Third, grounding tests: pure text models degrade markedly on questions that need visual attributes; synthetic tasks such as gSCAN and ReaSCAN show that language models without grounding cannot pass grounding generalization tests.

This paper's difference is only one point: it uses controlled ambiguity to make the "text limit" analytically computable, and splits the residual into "the irreducible floor determined by the modality" and "the reducible part eliminable by grounding/added data". The idea of measuring is not original to this paper; what this paper does is a controlled version whose floor can be computed analytically.

In addition, the symbol grounding problem (Harnad) points out that the connection between a symbol and its referent cannot be obtained from inside the symbols; this paper's formulation is "the text-only semantic residual has an analytic floor, and grounding can fill that floor in".

On evaluation, recent work has begun to report "the human ceiling", and this paper's approach can be seen as a controlled version of it: because we know the data generation process, we can compute the true upper bound rather than estimate it.

Inside the series: Paper 4 (structure for compute) shows that aligned corpora improve sample efficiency; Paper 10 (color as the first building block) shows that perceptual primitives can precede language alignment. This paper connects these two to "the semantic residual": what grounding lowers is the residual, not merely the efficiency.

VIII. Reproducibility

Script: 820_semantic_residual.py. Depends on Python 3 and numpy, no GPU, low memory footprint (under a hundred MB).

Appendix: Annotations

- **[definition]** Semantics = the alignment mapping between language and phenomena (with dimensions by modality).

- **[definition]** Semantic residual = the minimum achievable error under a given modality combination.

- **[measurable]** In the toy world, this residual can be computed analytically and matches the model (color 0.750/0.725/1.000, etc.).

- **[reducible]** After grounding the residual drops to 0 (the visual dimension of this experiment).

- **[undetermined]** The size of the real semantic residual; whether R-language is non-empty.

- **[not claimed]** It is not claimed that "semantics can be fully functionized", nor that it is "non-functionizable" — R is only declared, never unfolded.

> This paper was created by He Qi, with DeepSeek (AI) assisting in experiment design and code writing; the views and responsibility belong to the author.